

PRODUCT LITERATURE

HIRSUTIN TABLET 5MG

Each tablet contains:
Minoxidil 5 mg

Preservative:
Sodium Benzoate 0.1%w/w

Description
Odourless, white coloured round flat tablet with score on one side.

Pharmacodynamics
Pharmacotherapeutic group: Pyrimidine derivatives.

Mechanism of action
Minoxidil lowers the elevated systolic and diastolic blood pressure by decreasing peripheral vascular resistance via vasodilation. The smooth musculature of the resistance vessels must be regarded as the site of action for the relaxant effect of minoxidil. The active metabolite of minoxidil activates the ATP-modulated potassium (K+ ATP) channel causing K+ efflux, hyperpolarization, and smooth muscle relaxation.

Pharmacodynamic effects
Sympathetic reflexes mediated by baroreceptors secondarily increase heart rate and myocardial contractility, thereby increasing cardiac output. In addition, the plasma renin activity is increased via sympathetic nervous system stimulation, which results in an increased angiotensin II concentration with subsequent increased aldosterone secretion. In this way, the renal sodium excretion is reduced and extracellular volume increased. The pulmonary artery pressure may occasionally increase after the administration of minoxidil alone, but it decreases with the recommended concomitant therapy (beta-blocker plus diuretic).

Paediatric population
As severe hypertension requiring multi-drug therapy is uncommon in children, paediatric use of minoxidil was limited in the development programme and has remained so in published literature. Data available in children younger than 10 years of age is very limited; it involves approximately 40 patients, eight of whom were under one year of age.

Pharmacokinetics
Absorption and distribution
After oral administration in humans, at least 90% of minoxidil is absorbed in the gastrointestinal tract. Minoxidil is detected within 30 minutes in the plasma. Maximum plasma levels are reached 60 minutes after administration.
Based on results from a relative bioavailability study, tablet and oral solution formulations have similar values in area under the serum concentration-time curve (AUC), maximum serum concentration, time to reach maximum serum concentration (approximately 40 minutes), and the type of effect (antihypertensive). There is no accumulation following chronic administration of oral tablets compared with single dose.
Minoxidil is not bound to plasma proteins.
Minoxidil does not cross the blood-brain barrier.

Metabolism
At least 90% of the administered minoxidil is metabolized in the liver. The primary metabolite in humans is the minoxidil O-glucuronide. Some polar metabolites are also produced. The known metabolites have a weaker antihypertensive effect than the active ingredient itself.

Elimination
In humans, minoxidil plasma concentrations decrease with an average half-life of approximately 4 hours. However, the duration of action is over several days. Minoxidil and its metabolites are dialyzable. The renal clearance of minoxidil corresponds to the glomerular filtration rate. No substantial changes in the glomerular filtration rate and the renal plasma flow could be detected under minoxidil.

Paediatric population
No pharmacokinetic data regarding minoxidil in the paediatric population is currently available.

Hepatic impairment
No data are available.

Indication
Hypertension

Route of administration
Oral

Recommended Dose
ADULTS and CHILD above 12 years old: Initially 5 mg daily in single or divided doses (elderly 2.5 mg). May increase by 5 - 10 mg daily at intervals of 3 or more days until optimum control is achieved. Maximum 50 mg daily

Contraindication
Minoxidil is contra-indicated in patients with a phaeochromocytoma because it may stimulate secretion of catecholamines from the tumour through its antihypertensive action.
Hypersensitivity to the active substance or to any of the excipient.

Warning and Precautions
Salt and water retention
If used alone, minoxidil can cause a significant retention of salt and water leading to physical signs such as oedema, and to clinical deterioration of some patients with heart failure. Diuretic treatment alone, or in combination with restricted salt intake is, therefore, necessary for all patients taking minoxidil. Haemodilution may occur leading to temporary decrease in haematocrit, haemoglobin, and erythrocyte count (by approximately 7% initially which then recovers to pre-treatment levels). The patient’s bodyweight, fluid and electrolyte balance should be monitored for evidence of fluid retention. Salt and water retention in excess of 1 to 1.5 kg may diminish the effectiveness of minoxidil. Patients should, therefore, be carefully instructed about compliance with diuretic therapy and a detailed record of body weight should be maintained.
The product should be used with particular attention to maintenance of salt and water balance in patients with renal impairment, but who are not on dialysis.

Renal failure or dialysis patients
Those patients with renal failure or on haemodialysis may require smaller doses of minoxidil.

Myocardial infarction
Patients who have had myocardial infarction should only be treated with minoxidil after a stable post-infarction state has been established.

Tachycardia
Because minoxidil is a vasodilator, reflex tachycardia may occur and possibly angina pectoris may occur in patients at risk; it is recommended that minoxidil be used in combination with beta-adrenergic blocking agent or other sympathetic nervous system suppressants to blunt or prevent such a response.

Hypertrichosis
Hypertrichosis occurs in most patients treated with minoxidil and all patients should be warned of this possibility before starting therapy. Most patients will experience an elongation, thickening and enhanced pigmentation of fine body hair. Usually these signs will emerge 3 to 6 weeks after starting treatment. They initially emerge in the face, and they may slightly subside with continued treatment. However, hypertrichosis was hardly or not at all tolerable in less than 10% of patients. Spontaneous reversal to the pre-treatment state can be expected one to six months after cessation of therapy.

ECG alterations
Soon after starting minoxidil therapy approximately 60% of patients exhibit ECG alterations in the direction and magnitude of their T waves. Large changes may encroach on the ST segment, unaccompanied by evidence of ischaemia. These asymptomatic changes usually disappear with continuing minoxidil treatment. The ECG reverts to the pre-treatment state when minoxidil is discontinued.

Thrombocytopenia and leucopenia
Thrombocytopenia and leucopenia have been rarely reported.

Pericarditis, Pericardial Effusion and Tamponade
Pericarditis, Pericardial Effusion and Tamponade – Although there is no evidence of a causal relationship, there have been multiple reports of pericarditis occurring in association with minoxidil. Pericardial effusion and occasionally tamponade, has been observed in about 3% - 5% of treated patients not on dialysis. While in many cases, the pericardial effusion is associated with other potential aetiologies, there have been cases in which these potential causes of effusion were not present. Patients should be observed closely for any suggestion of a pericardial effusion and pericardiocentesis, or surgery may be required. If the effusion persists, withdrawal of minoxidil should be considered in light of other means of controlling the hypertension and the patient’s clinical status.

Galactose intolerance
Patients with rare hereditary problems of galactose intolerance, the Lapp lactose deficiency or glucose-galactose malabsorption should not take this medicine.

Paediatric population
Children strictly require appropriate and individualised dosing of minoxidil, betablockers and diuretics. They should be under close specialist supervision in hospital. Caution is required when there is significant renal impairment. The development of peripheral oedema or any signs suggestive of congestive heart failure or of pericardial or pleural effusion should be carefully watched for. Renal function should be monitored. Body weight and urine output should be monitored. Regular follow up must be ensured during treatment with minoxidil. Before starting treatment parents and carers should be warned of the likely occurrence of hypertrichosis.

Interactions with Other Medicament
The effect of minoxidil may be additive to concurrent antihypertensive agents and other agents with blood pressure lowering effects. The interaction of minoxidil with sympathetic-blocking agents such as guanethidine or betanidine may produce excessive blood pressure reduction and/or orthostatic hypotension. If possible guanethidine should be discontinued well before minoxidil is begun. If this is not feasible, minoxidil therapy should be instituted in the hospital and the patient monitored carefully for orthostatic events.

Pregnancy and Lactation
Pregnancy
There is limited data from the use of minoxidil in pregnant women. Studies in animals have shown reproductive toxicity. Minoxidil is not recommended during pregnancy and in women of childbearing potential not using contraception. Neonatal hirsutism has been reported following exposure of minoxidil during pregnancy.

Breast-feeding
Minoxidil has been reported to be excreted in human milk. A risk to the suckling child cannot be excluded. A decision must be made whether to discontinue breastfeeding or to discontinue/abstain from minoxidil therapy taking into account the benefit of breast-feeding for the child and the benefit of therapy for the woman.

Fertility
There are no fertility data from the use of minoxidil in humans. In a fertility study with male and female rats, a dose-dependent reduction of the conception rate was found. The dose corresponded to one to five times the maximum dose used in humans to treat hypertension.

Side Effects
Most patients receiving minoxidil experience a diminution of pre-existing side-effects attributable to their disease or previous therapy.

MedDRA System Organ Class	Frequency	Undesirable Effects
Blood and Lymphatic System Disorders	Rare	Leucopenia, thrombocytopenia
Metabolism and Nutrition Disorders	Common	Fluid retention, oedema
Cardiac Disorders	Very Common	Tachycardia, pericarditis
	Common	Pericardial effusion, cardiac tamponade
	Not Known	Angina pectoris
Respiratory, Thoracic and Mediastinal Disorders	Not Known	Pleural effusion
Gastrointestinal Disorders	Not Known	Gastrointestinal disorder
Skin and Subcutaneous Tissue Disorders	Very Common	Hypertrichosis, hair colour changes
	Rare	Stevens-Johnson syndrome, dermatitis bullous, rash,
Reproductive System and Breast Disorders	Not Known	Breast tenderness
General Disorders and Administration Site Conditions	Not Known	Peripheral oedema associated with or independent of weight gain
Investigations	Very Common	ECG abnormal
	Not Known	Blood creatinine increased Blood urea increased

***ADR:** Hirsutism, Rash Maculo-Papular, Hair Growth abnormal, Hirsutism Face, Hirsutism

Symptoms and Treatment of Overdose
If exaggerated hypotension is encountered, it is most likely to occur in association with residual sympathetic nervous system blockade (guanethidine-like effects or alpha-adrenergic blockade). Recommended treatment is intravenous administration of normal saline. Sympathomimetic drugs, such as noradrenaline (norepinephrine) or adrenaline (epinephrine), should be avoided because of their excessive cardiac-stimulating action. Phenylephrine, angiotensin II and vasopressin, which reverse the effect of minoxidil, should be used only if inadequate perfusion of a vital organ is evident.

Effects on Ability to Drive and Use Machine

No studies on the effect of minoxidil on the ability to drive or use machines have been performed. The ability to drive or operate machinery may be influenced by the individual response to treatment, particularly at the start of therapy.

Packing

Packed in blister-packs of 10 x 10’s/box.

Storage Conditions

Store below 30°C in a dry place, protected from direct light.
Keep away from reach of children.

Expiry Date

3 years from date of manufacture

Name and Address of Manufacturer:

TERAPUTICS SDN. BHD. (590500-W)
Lot 10 & 11, PERDA Industrial Park,
Lorong IKS Simpang Ampat B,
14100 Simpang Ampat, S.P.S.,
Pulau Pinang, Malaysia

Name and Address of Product Registration Holder/ Distributor:

ZONTRON PHARMACEUTICALS SDN.BHD. (445695-T)
Lot 10 & 11, PERDA Industrial Park,
Lorong IKS Simpang Ampat B,
14100 Simpang Ampat, S.P.S.,
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Malaysian Drug Registration No. :

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